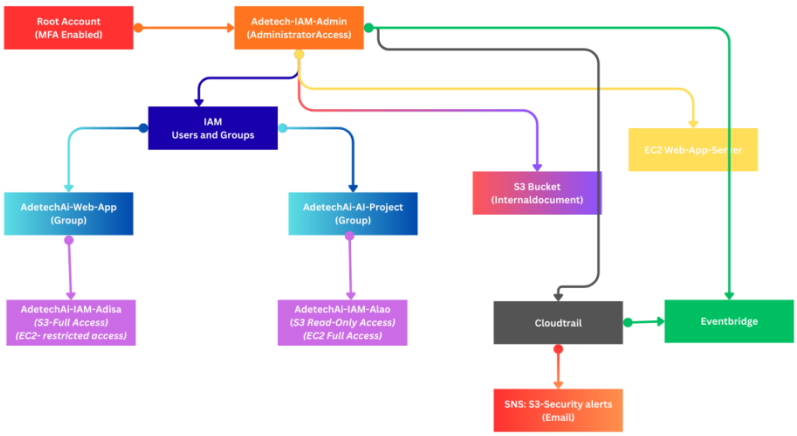
Naztech Cloud Security Implementation Project
(ISO 27001 / NIST / CIS Aligned)

1. Executive Summary

This project demonstrates the implementation of foundational cloud security controls aligned with ISO/IEC 27001:2022, NIST Cyber security Framework (CSF), and CIS Critical Security Controls. The objective was to establish a secure AWS environment with strong identity governance, access control, logging, monitoring, and alerting mechanisms.

The environment was designed to:

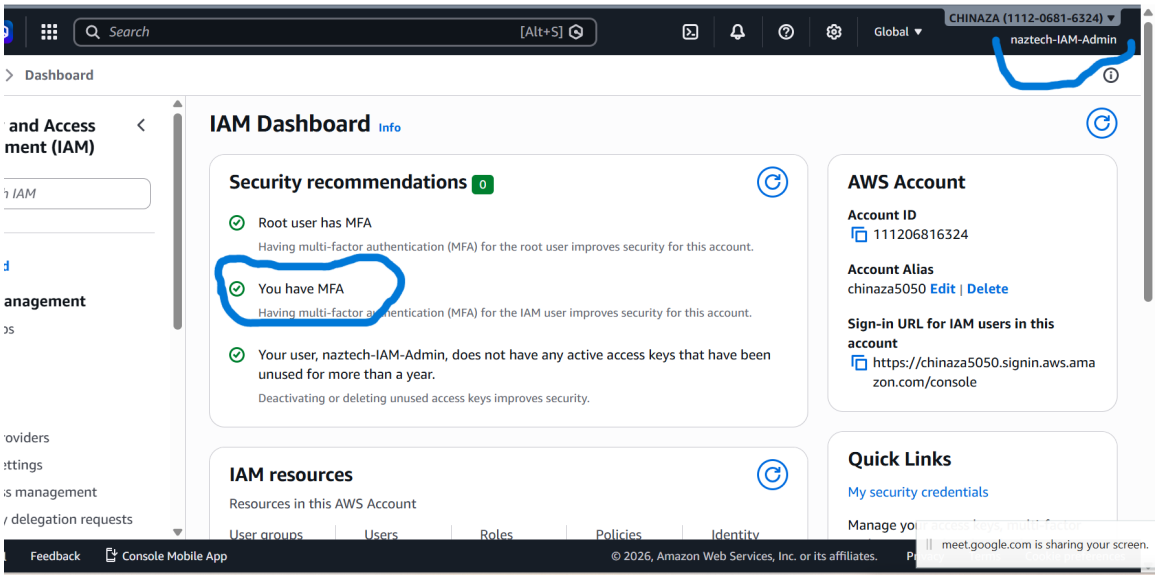
- Secure administrative access
- Enforce separation of duties
- Protect sensitive S3 resources
- Restrict EC2 administrative actions
- Capture and audit API activity
- Generate automated alerts for sensitive S3 actions



2. Governance and Account Security

ISO 27001 A.5 & A.6 | NIST CSF ID.GV | CIS Control 1

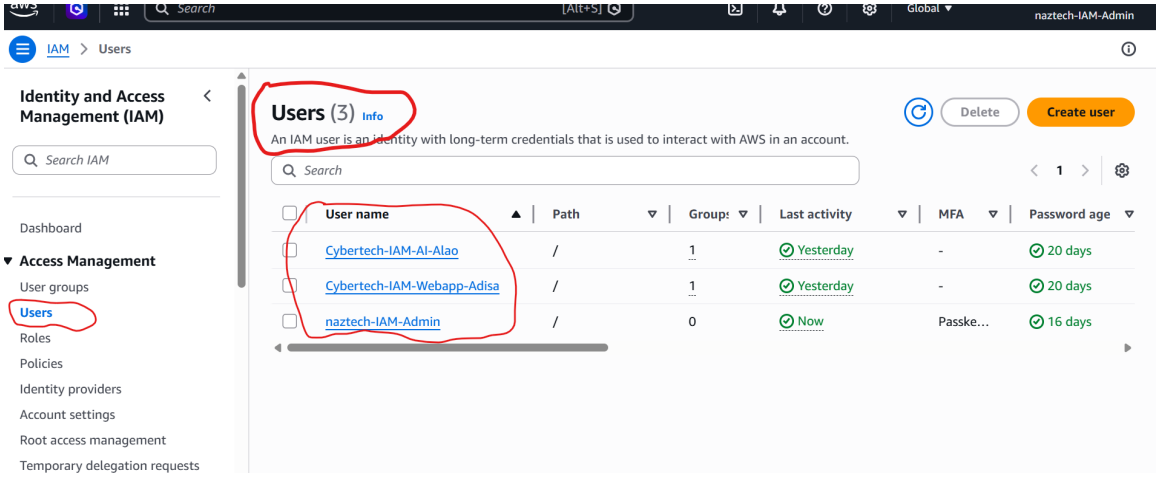
The AWS root account was secured with Multi-Factor Authentication (MFA) and restricted from daily use. In accordance with governance best practices, a dedicated administrative IAM user (naztech-IAM-Admin) was created to handle operational activities, ensuring accountability and reducing single-point-of-failure risk.

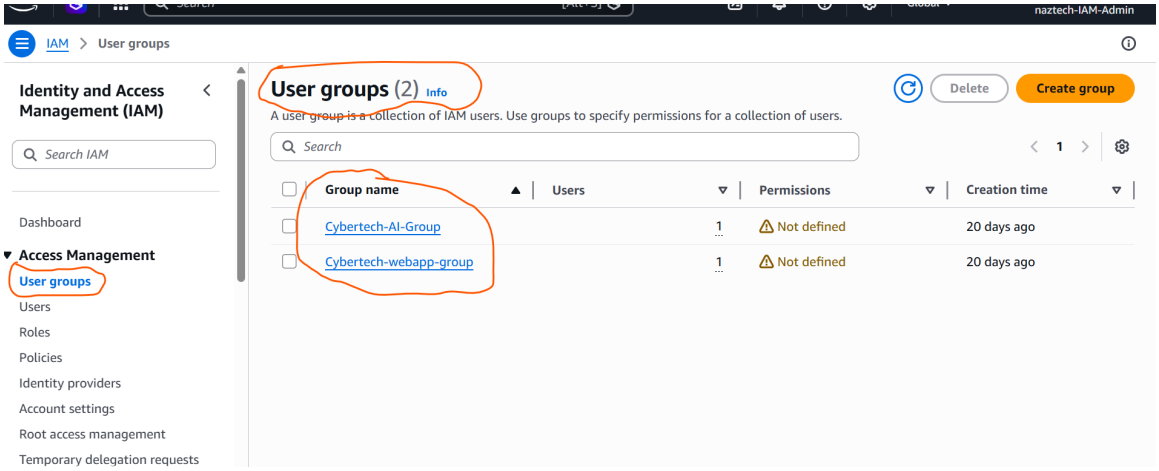


3. Identity and Access Management (IAM)

ISO 27001 A.5.15, A.8 | NIST CSF PR.AC | CIS Control 5

IAM users and groups were created to enforce role-based access control (RBAC). Users were assigned to groups based on job function, ensuring separation of duties and least privilege.

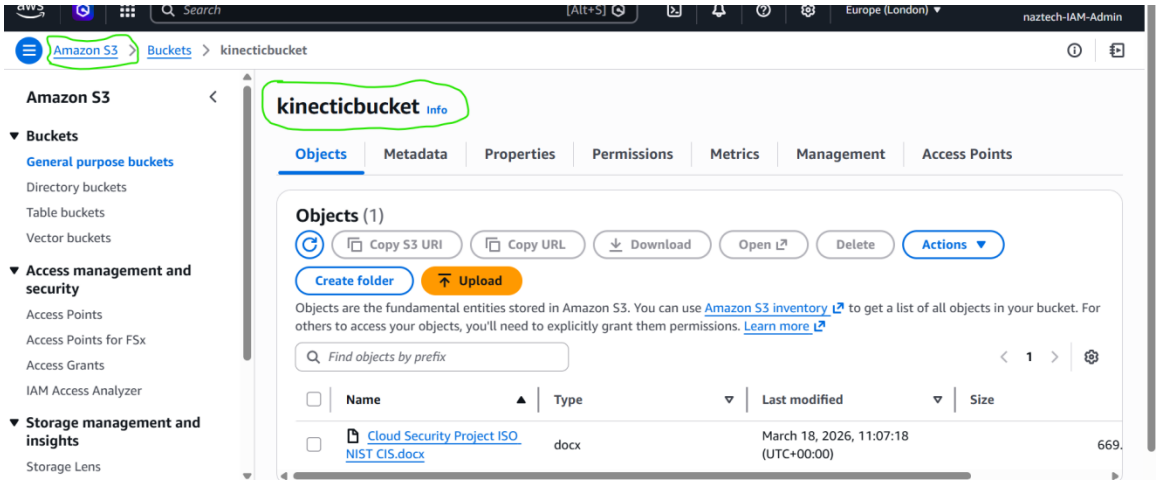




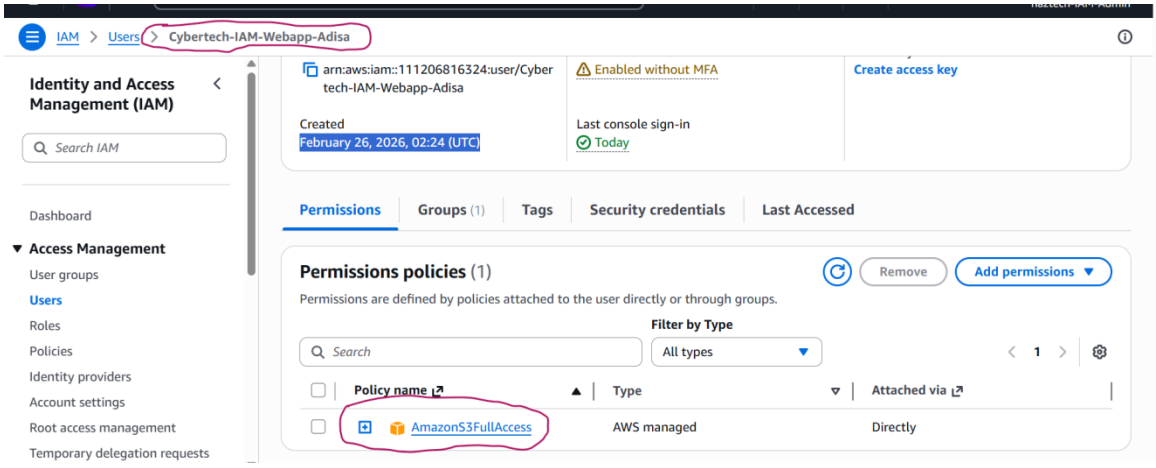
4. Object Storage Security (Amazon S3)

ISO 27001 A.8.2 | NIST CSF PR.DS | CIS Control 3

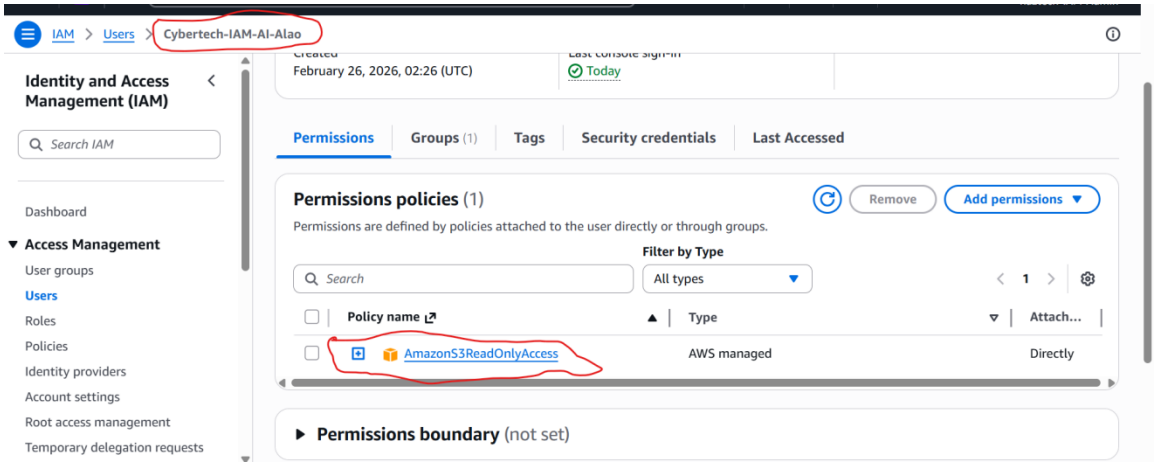
An S3 bucket containing kinetic bucket was created and protected using IAM-based RBAC. One user was granted full S3 access while another was restricted to read-only permissions. Access validation confirmed enforcement of least privilege.



cybertech-IAM-webapp-Adisa Fullaccess



cybertech-IAM-AI-Alao Readonlyaccess



5. Compute Resource Access Control (EC2)

ISO 27001 A.8.9 | NIST CSF PR.AC-4 | CIS Control 4

An EC2 instance was deployed to simulate a web application server. Inline IAM policies were used to explicitly deny high-risk administrative actions such as instance termination and key pair creation for selected users. Policy enforcement was verified through controlled testing.

6. Logging and Monitoring

ISO 27001 A.8.15 | NIST CSF DE.CM | CIS Control 8

AWS CloudTrail was configured to capture management events across all IAM users. Centralized logging ensures traceability, supports incident response, and enables compliance auditing.

7. Security Event Detection and Alerting

ISO 27001 A.8.16 | NIST CSF DE.AE | CIS Control 8

EventBridge rules were created to detect sensitive S3 actions captured by CloudTrail. Detected events trigger notifications via Amazon SNS, delivering near real-time alerts to security personnel.

8. Conclusion

The project demonstrates practical application of internationally recognized security frameworks

in a cloud environment. Controls implemented align with governance, protection, detection, and response requirements, making the environment audit-ready and suitable for enterprise use.